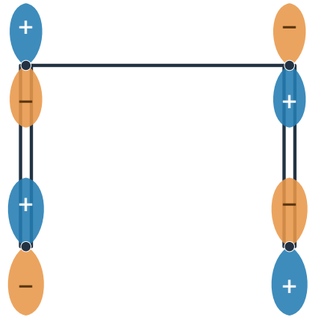
4π electrocyclic closure | orbital phases along the reaction

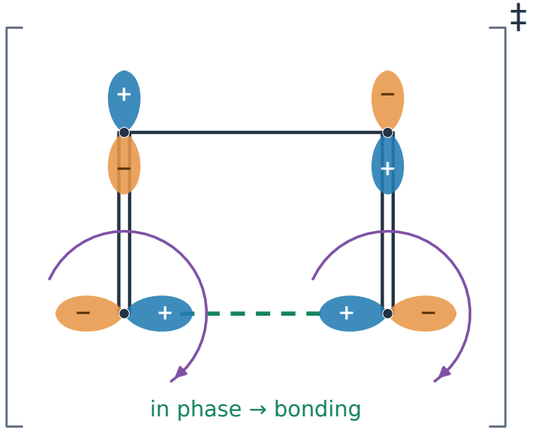
THERMALLY ALLOWED

REACTANT ORBITALS	PROPOSED TRANSITION STATE	PRODUCT
Polyene HOMO, ψ₂ (occupied)	conrotatory Möbius circuit · 4 electrons	Product connectivity



Opposite terminal phases

Δ



Δ

Cyclobutene

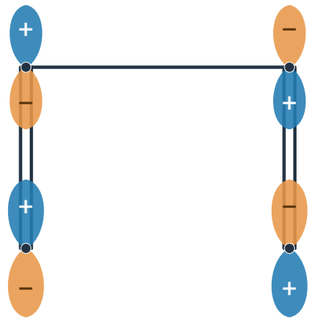
Follow the terminal lobes during rotation.

same rotation sense

New σ bond(s) highlighted in green.

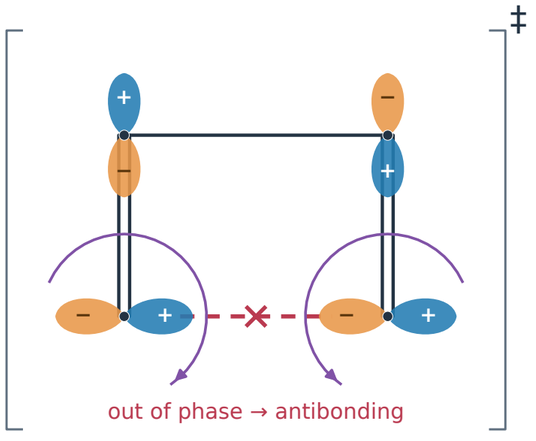
THERMALLY FORBIDDEN

Polyene HOMO, ψ₂ (occupied)	disrotatory Hückel circuit · 4 electrons	Proposed product connectivity



Opposite terminal phases

Δ



Cyclobutene

Follow the terminal lobes during rotation.

opposite rotation senses

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Rotation arrows follow the same viewing convention at both termini. Product connectivity is shown without substituent stereochemistry.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.